

School of Computing & IT
Web Programming (CIT2011)
Week 1 – Exploring the Web with Developer Tools

1. Lab Overview

Modern web developers need to understand what happens inside the browser when a webpage loads and when a user interacts with it. In this lab, students use Chrome, or Firefox Developer Tools to inspect HTML, and CSS.

Register to <https://onecompiler.com/>. You will use this online compiler to share your ongoing labs with your tutor for review.

2. Learning Outcomes

1. Open and navigate browser Developer Tools.
2. Identify HTML elements using the Elements/Inspector panel.
3. Modify HTML and CSS temporarily using Developer Tools.
4. Use responsive/device emulation to test different screen sizes.
5. Use Developer Tools as part of a modern web-development workflow.

3. Software Requirements

Students need the following:

- Google Chrome / Mozilla Firefox
- Internet connection
- A text editor such as VS Code

4. Developer Tools Overview

Chrome	Firefox	Purpose
Elements	Inspector	Examine HTML and CSS
Console	Console	Execute JavaScript and view errors
Sources	Debugger	Examine JavaScript/source files
Network	Network	Examine HTTP requests
Application	Storage	Examine cookies/storage
Lighthouse	Lighthouse	Performance/accessibility analysis
Device Toolbar	Responsive Design Mode	Test different screen sizes

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PART A — Opening Developer Tools

Activity 1: Open Developer Tools

- a. Open <https://www.w3schools.com/html/default.asp> or another publicly accessible website of your choice).
- b. Open Developer Tools using right-click → Inspect, F12, Ctrl+Shift+I on Windows, or Cmd+Option+I on macOS.

Question 1. What is the name of the panel used to inspect the webpage's HTML? Elements Panel

Question 2. List five panels available in Developer Tools.

1. Elements 2. Console 3. Sources
4. Lighthouse 5. Network

PART B — Inspecting HTML

Activity 2: Examine the HTML Structure

Select the Elements/Inspector panel. Locate <html>, <head>, <body>, headings, paragraphs, links, images and navigation elements.

Question 3. Identify one heading element. Record the HTML tag and the text inside it.

HTML tag: <h2> </h2>

Text: Learning By Examples

Question 4. Find one hyperlink. Record the HTML element and its href value.

HTML element: <a>

href: html_editors.asp

Activity 3: Modify HTML

Find a visible heading. Right-click the element and select Edit as HTML (or double-click it). Temporarily change it to:

<h1>CIT2011 Developer Tools Lab</h1>

Question 5. What happened to the webpage?

The words CIT2011 Developer Tools Lab replaced the heading that I updated in the HTML.

Question 6. Was the original website permanently changed? Explain why.

No it was not permanently changed, this is because the edits I made were only to the local copy of the website on my browser and not to the files on the website's server.

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PART C — Inspecting CSS

Activity 4: Examine CSS

Select an HTML element containing visible text. Examine the Styles panel and identify CSS properties controlling font size, font family, margin, padding, text colour and background colour.

Property	Value
font-size	32px;
font-family	"Segoe UI", Arial, sans-serif;
margin	10px 0;
padding	0;
color	#0000;
background-color	#fff;

Activity 5: Modify CSS

Add the following declaration to a heading in the Styles

panel: font-size: 50px;

Then change it to:

font-size: 20px;

Question 1. What happened when the font-size value changed?

The heading text became smaller when the value changed from 50px to 20px.

Question 2. Add text-transform: uppercase; What happened?

All the letters became uppercased.

Question 3. Change margin and padding. Explain the difference between margin and padding.

Margin controls the space outside the element to determine how far other elements should be away from the element, while padding deals with the internal spacing between the content and its border.

PART D — Understanding the Box Model

Activity 6: Inspect the Box Model

Select an element and locate the Box Model diagram.

Question 4. Identify the four major components of the CSS box model.

1. Content 2. Border 3. Padding 4. Margin

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PART I — Responsive Web Design

Activity 15: Device Emulation

Open Device Toolbar / Responsive Design Mode and test mobile, tablet, laptop and desktop sizes such as 375×667, 768×1024 and 1366×768.

Question 5. What happened to the webpage when the screen width became smaller?

The elements on the page were resized and adjusted to fit the screen type.

Question 6. Did the navigation change? Describe what happened.

The screen became scrollable, and many of the options were moved to a hamburger menu at the top of the screen to avoid clustering and clashing elements.

Question 7. Did images resize? Explain your observation.

Yes, the images resized themselves and scaled down to fit the smaller screen without overflowing. They adjusted proportionally so the layout stayed clean and no horizontal scrolling was needed.

Activity 16: Test Your Own Responsive Design

Create a file named responsive.html using the following code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Responsive Test</title>
  <style>
    body { font-family: Arial; margin: 20px; }
    .container { display: flex; gap: 20px; }
    .box { padding: 30px; background-color: lightblue; flex: 1; }
    @media (max-width: 600px) {
      .container { flex-direction: column; }
    }
  </style>
</head>
<body>
  <h1>Responsive Web Design</h1>
  <div class="container">
    <div class="box">Box 1</div>
    <div class="box">Box 2</div>
    <div class="box">Box 3</div>
  </div>
</body>
</html>
```

Use Developer Tools to test 1200px, 900px, 700px, 600px, 500px and 375px.

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Question 8. At approximately what width does the layout change from rows to columns?

600px

Question 9. Which CSS feature caused the layout to change?

@media

Question 10. Complete the table above. (Unsure what table miss)

FINAL CHALLENGE — Activity 20: Become the Web Detective

A student reports that a webpage is displaying incorrectly on their mobile phone, although it works correctly on a laptop. Use Developer Tools to investigate responsive layout, CSS rules, media queries, images, JavaScript errors, network requests and HTTP status codes.

Question 11. Identify three possible reasons why a webpage may work correctly on desktop but incorrectly on mobile.

1. Because the website was only formatted for desktop screens and not mobile screens; @media not used.
2. Overflow was not accounted for when designing the CSS, so images or other elements could have fixed widths from the use of px.
3. The viewport meta tag is missing, so the phone renders the page at desktop width instead of fitting the screen.

FINAL REFLECTION

Question 12. How does Developer Tools help a web developer understand a webpage?

Developer Tools allows developers to view and edit the HTML, CSS and JavaScript behind a page in real time, which lets them see each part of the webpage and understand how each works.

Question 13. Why is it useful to inspect HTML and CSS rather than simply looking at the webpage?

Looking at the webpage does not show the underlying HTML and CSS on a page, whereas inspecting shows all the elements and styling, giving better understanding to how everything works.

Question 14. Why is the Network panel important to a web developer?

The Network panel shows every file and request on the page and the HTTP status of each one, allowing developers to accurately track them and spot slow, loading or failed files/requests.

Question 15. Why is responsive testing important?

Responsive testing is important because users view websites on many different screen sizes, and the design on one screen size may not work on another. Therefore testing helps see what needs to be edited to make the proper

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adjustments on relevant screen sizes.

Question 16. How can Developer Tools help identify bugs?

Developer Tools helps identify bugs by showing JavaScript errors in the Console, broken styles in the Styles panel, failed requests in the Network panel, and layout problems through device emulation.

Question 17. Explain how HTML, CSS and JavaScript work together to create an interactive webpage.

HTML provides the structure and content. CSS controls the appearance and layout. JavaScript adds interactivity and dynamic behaviour. When the three are combined, the webpage becomes content filled, styled and interactive for users.

Question 18 – Build a webpage

Use the template provided by your tutor to build a webpage using HTML5 and basic CSS3. Webpage must include, basic HTML5 tags such as, image, paragraph, navigation, heading, sections, footer, links, etc.

Practical Deliverables

- Completed lab sheet (Questions 1–18).
- At least six screenshots: HTML inspection, CSS inspection
- Responsive.html

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Screenshots

The screenshot displays the W3Schools HTML Tutorial page in a web browser. The page features a navigation bar with links to various programming topics like HTML, CSS, JavaScript, SQL, Python, Java, PHP, C, C++, C#, AWS, W3.CSS, and HOW TO. The main content area is titled "HTML Tutorial" and includes a "Learn HTML" section with a description of HTML as a standard markup language for web pages. Below this, there's an "Adventure Mode" section promoting a new W3Schools Adventure App available on the App Store and Google Play. A progress bar indicates the user's progress through the tutorial. The "Learning by Examples" section mentions a "Try it Yourself" editor. On the right side, the browser's developer tools are open, showing the "Elements" panel with the DOM tree, the "Styles" panel with CSS rules, and the "Console" panel with network requests. A box model diagram is also visible in the Styles panel.

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HTML Tutorial

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Learn HTML

HTML is the standard markup language for Web pages.

With HTML you can create your own Website.

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Learn HTML in bite-sized quests with XP and streaks. Your progress grows for each new adventure! Free on iOS and Android.

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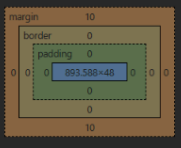
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SHARE

With our "Try it Yourself" editor, you can edit the HTML code and view the result in the browser:

Elements Console Sources +

text for it and falls the page with ASP 0238 before it ever renders. -->
<div class="ws-share ws-share--float"></div> (flex)
<script></script>
<h2>CIT2011 Developer Tools Lab</h2> --> 50 QP
<p></p>
<div class="w3-example"></div>
<div></div>

Styles Computed Layout Event Listeners DOM breakpoints Properties >
Filter (x) show .cls +
Pseudo :before element
, *:before, *:after {
box-sizing: inherit;
}
Pseudo :after element
, *:before, *:after {
box-sizing: inherit;
}


Console Issues +
Filter
▶ GET https://sync.sparteo.com/crossfire.js fast-cmp-stub.js:1
net::ERR_BLOCKED_BY_CLIENT
▶ POST https://t.fastcmp.com/beacon fast-cmp-stub.modern.28aef23199775a.js:1
net::ERR_BLOCKED_BY_CLIENT
▶ POST https://t.fastcmp.com/beacon fast-cmp-stub.modern.28aef23199775a.js:1
net::ERR_BLOCKED_BY_CLIENT

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<p></p>

Styles Computed Layout Event Listeners DOM breakpoints Properties >
Filter (x) show .cls +
element.style {
}
h2 {
font-size: 50px;
}
h1, h2, h3, h4, h5, h6 {
font-family: "Segoe UI", Arial, sans-serif;
font-weight: 400;
margin: 10px 0;
}
h2 {
font-size: 30px;
}
, *:before, *:after {
box-sizing: inherit;
}
h2 {
display: block;
font-size: 1.5em;
margin-block-start: 0.83em;
margin-block-end: 0.83em;
}

Console Issues +
Filter
▶ GET https://sync.sparteo.com/crossfire.js fast-cmp-stub.js:1
net::ERR_BLOCKED_BY_CLIENT
▶ POST https://t.fastcmp.com/beacon fast-cmp-stub.modern.28aef23199775a.js:1
net::ERR_BLOCKED_BY_CLIENT
▶ POST https://t.fastcmp.com/beacon fast-cmp-stub.modern.28aef23199775a.js:1
net::ERR_BLOCKED_BY_CLIENT

Example

```
<!DOCTYPE html>  
<html>  
<head>
```

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The screenshot shows the W3Schools website with the 'Learn HTML' section. The page includes a navigation bar with links to various programming topics. The main content area features a 'Learn HTML' section with a description of HTML as a standard markup language and a 'Try it Yourself' editor. The 'Adventure Mode' section is also visible, promoting a game-like learning experience. The developer tools are open on the right, showing the 'Elements' panel with the HTML structure of the page, the 'Styles' panel with the default styles for the selected element, and the 'Console' panel with network requests.

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Learning by Examples

With our "Try it Yourself" editor, you can edit the HTML code and view the result in the browser:

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
```

The screenshot shows the W3Schools website with the 'Learn HTML' section. The page includes a navigation bar with links to various programming topics. The main content area features a 'Learn HTML' section with a description of HTML as a standard markup language and a 'Try it Yourself' editor. The 'Adventure Mode' section is also visible, promoting a game-like learning experience. The developer tools are open on the right, showing the 'Elements' panel with the HTML structure of the page, the 'Styles' panel with the default styles for the selected element, and the 'Console' panel with network requests.

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CIT2011 DEVELOPER TOOLS LAB

With our "Try it Yourself" editor, you can edit the HTML code and view the result in the browser:

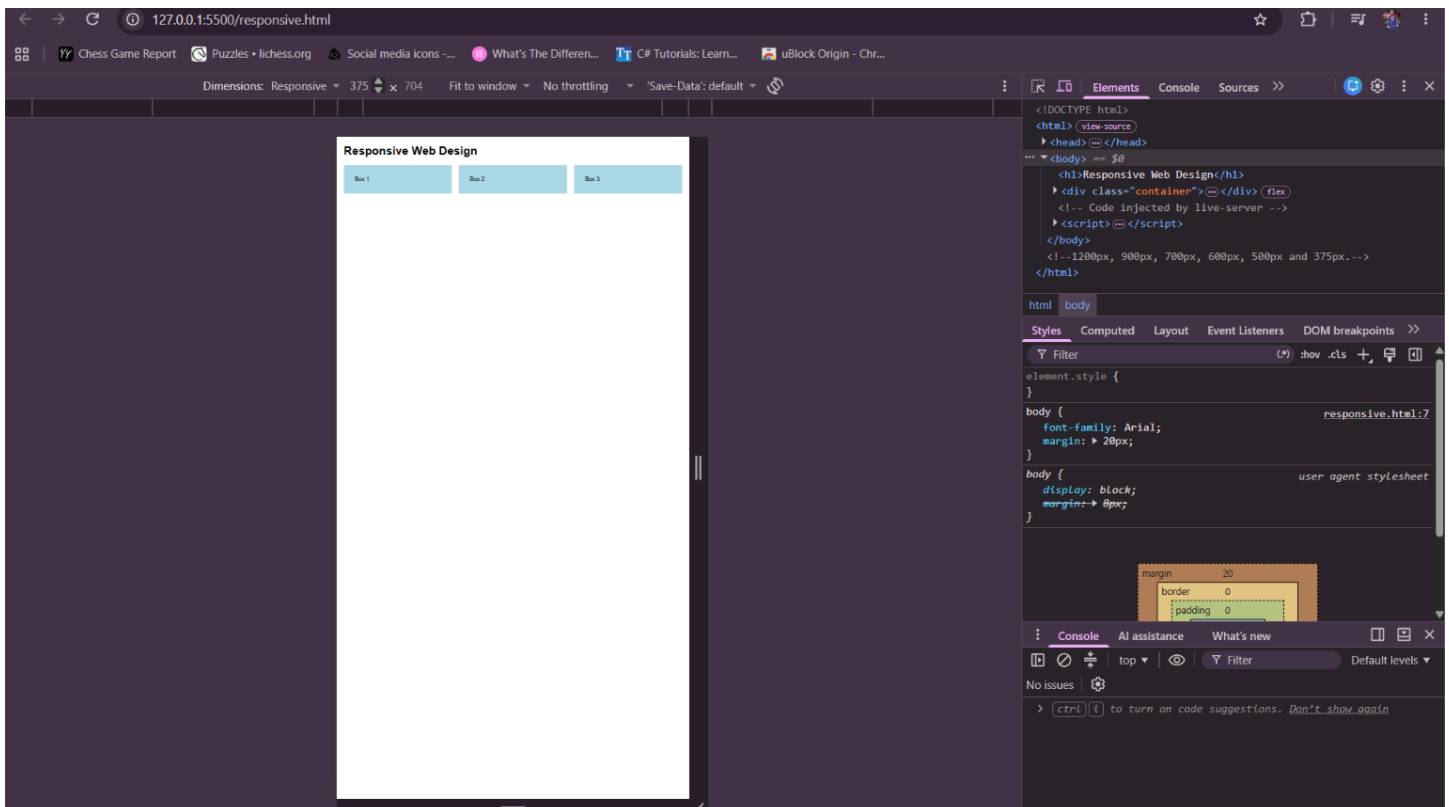
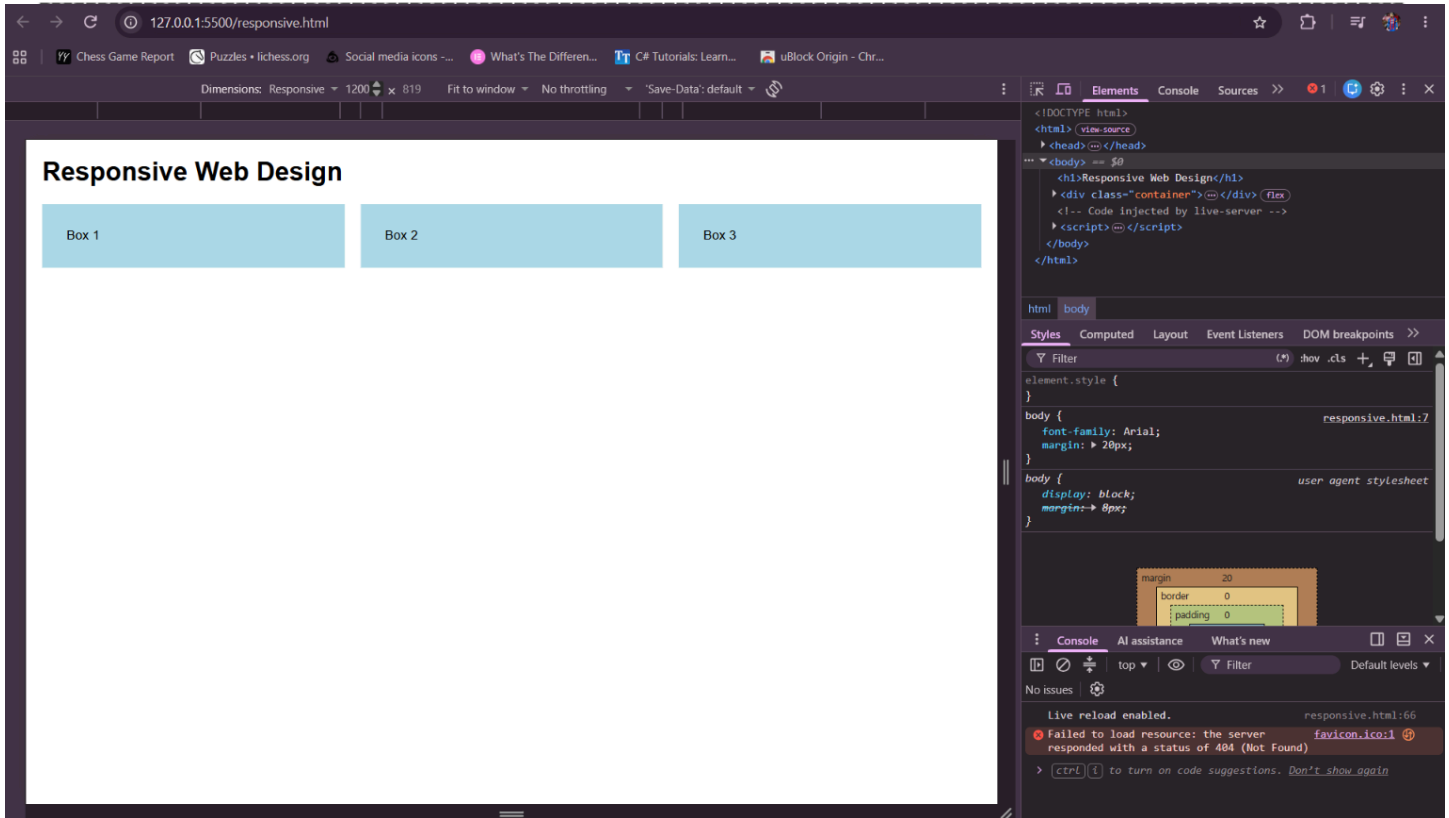
Example

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
```

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